



11.3 Particle -wave duality of matter



De Broglie hypothesis

□ 1923, French physicist **Victor Pierre De Broglie** suggested in his Phd thesis .



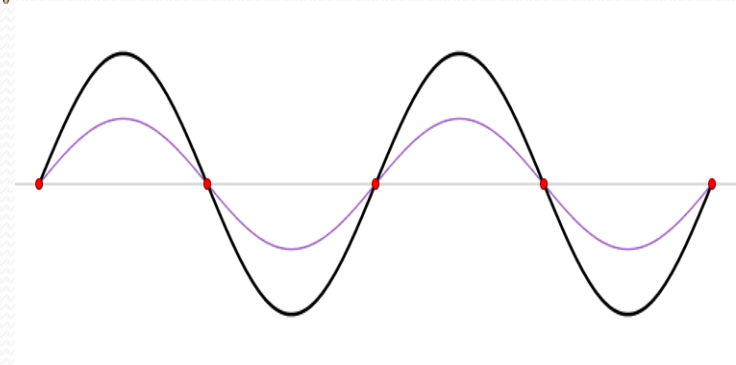
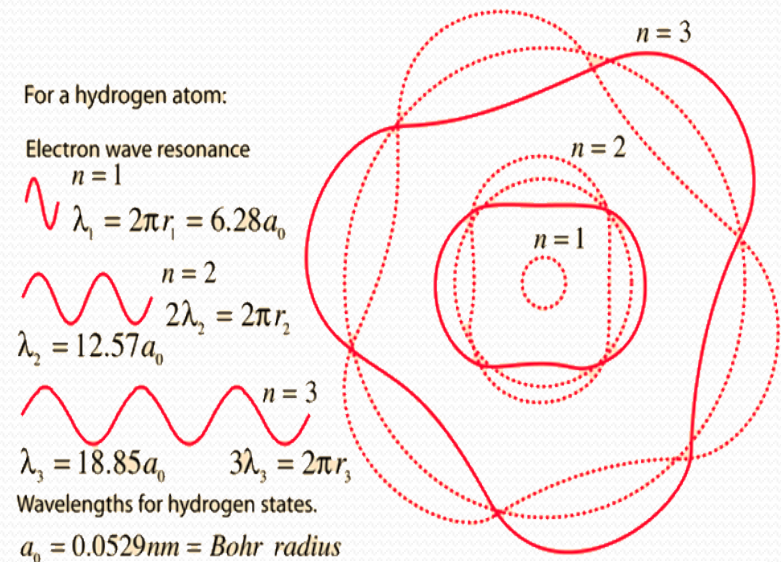
Louis de Broglie
(1892-1987)

- Matter has wave properties in the same way waves have matter properties
- Every particle shows wave properties but tiny particles like electrons has much wave-like nature



Given rise to explain quantized states of Bohr Model

- ❑ Electrons has wave properties
- ❑ Electrons are standing waves around the nucleus
- ❑ Each wave has different wave lengths





De Broglie Wavelength

Einstein's special relativity

$$P = E/C$$

But , $E = hf$, $C = f \lambda \longrightarrow E = hc / \lambda$

$$\lambda = h/p;$$

$$\lambda = h/mu$$

- For all matter,

$$\text{Wave length} \propto 1/\text{Momentum}$$

λ - De Broglie Wave length

h - Plank constant

P - Momentum

C - Velocity of light

$$h = 6.624 \times 10^{-34}$$



$$\lambda = h/mu$$

❑ Macroscopic Objects

Mass ↑ **Momentum** ↑ Wave length ↓

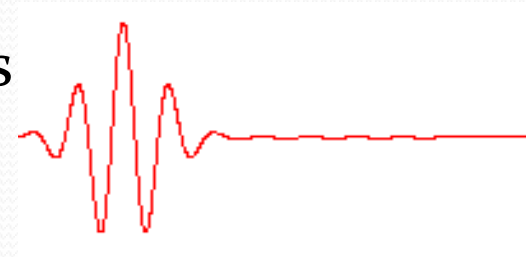
Minimum wave properties



❑ For microscopic particles Neutrons, Protons

Mass ↓ Momentum ↓ **Wave length** ↑

Maximum Wave properties



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Energy & De Broglie wave Length

De Broglie wavelength $\lambda = h/p$,
 $\lambda = h/mu$

$$E = (1/2) mu^2$$

$$2E = mu^2$$

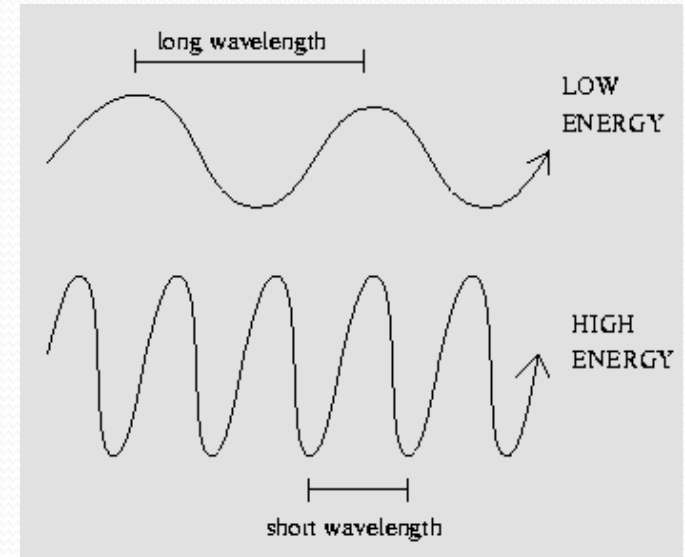
$$2Em = m^2u^2$$

$$2Em = (p)^2$$

$$E = P^2/2m$$

$$\lambda = h/\sqrt{2Em}$$

□ When E (Kinetic Energy)
increases λ decreases

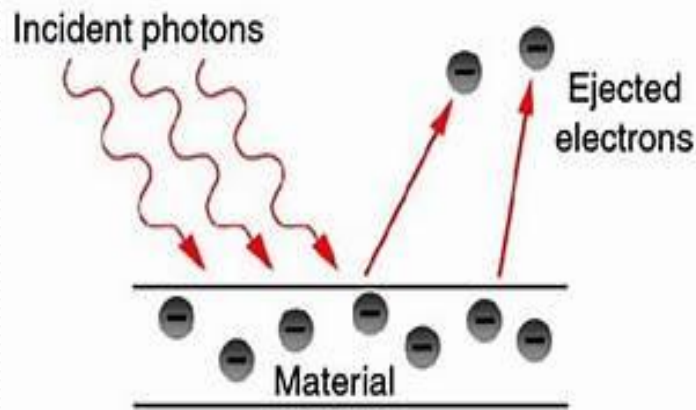




Particle Duality

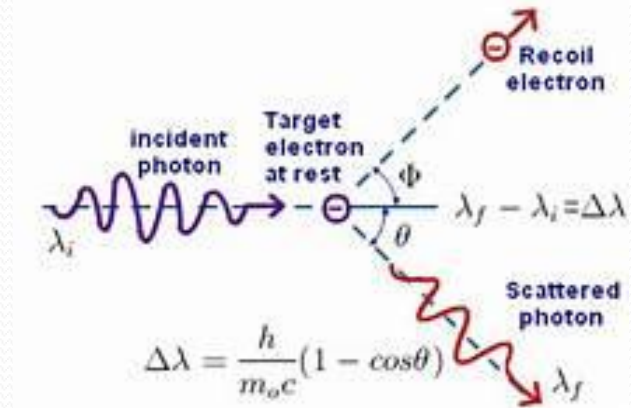
□ Particlelike nature of waves

(Photo Electric Effect)



□ Photons incident on metal surfaces emit electrons

(Compton Effect)



□ Photons collision with electrons results new photons with moving electrons.



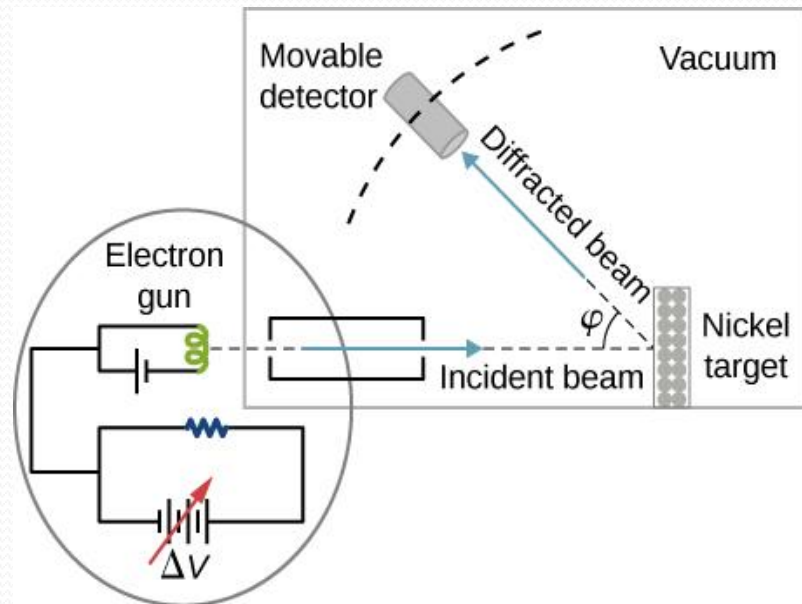
Wavelike nature of particle

❑ Electron diffraction

- 1924, Discovered by Dawison and Germer
- This experiment was a breaking background for Quantum mechanics .

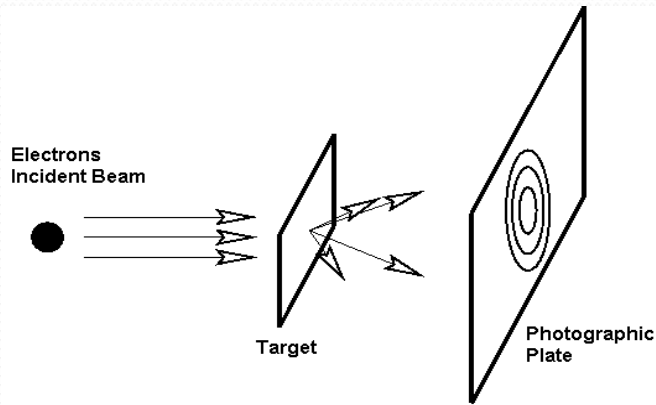


- De Broglie hypothesis was experimentally proved true

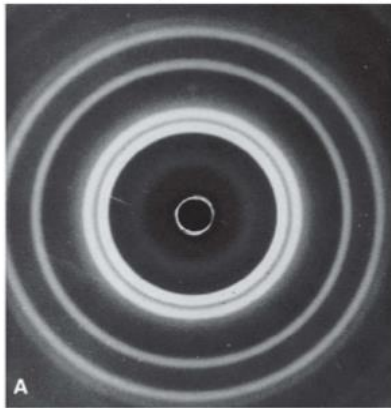




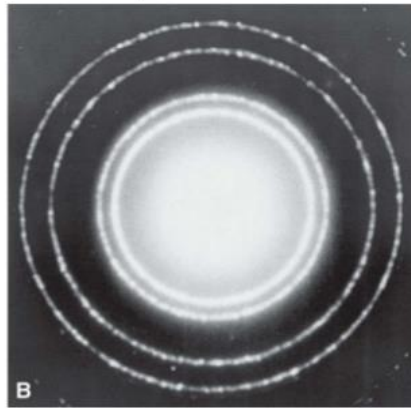
J.P Thompson exposed Al powder to X rays and Electrons and received diffraction patterns



Comparing the diffraction patterns of x-rays and electrons.



x-ray diffraction of aluminum foil



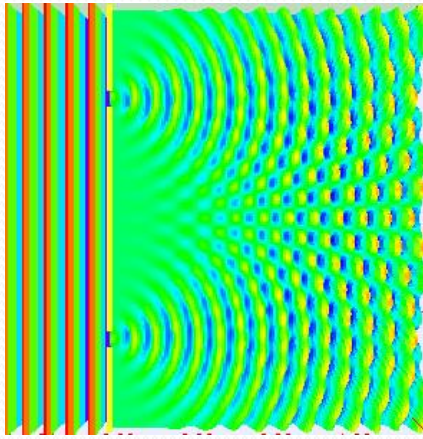
electron diffraction of aluminum foil

❑ Calculated wavelength
Matched with the De
Broglie wavelengths

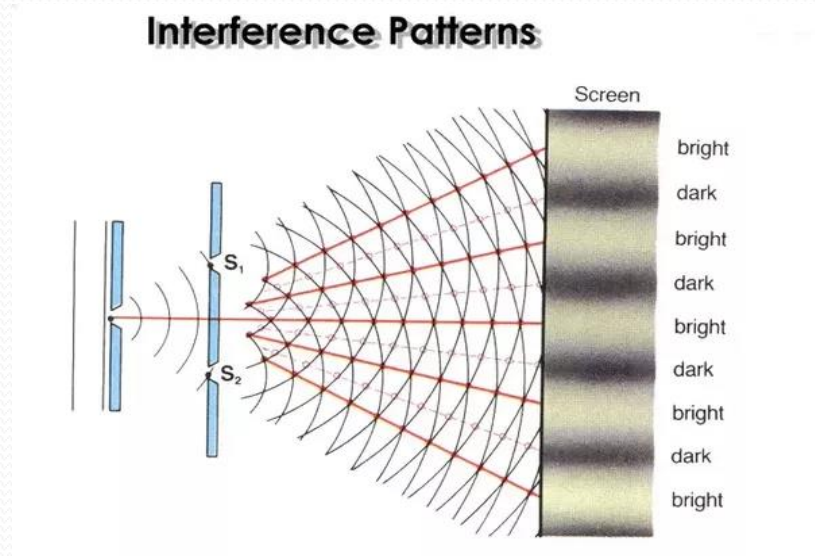


Matter and Wave interference

- Young's Double slit Experiment



□ Super imposed waves create interference fringes

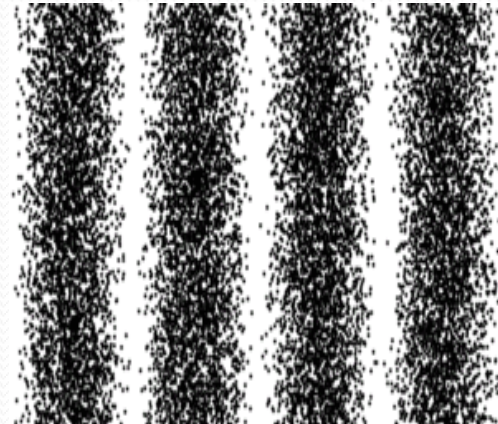
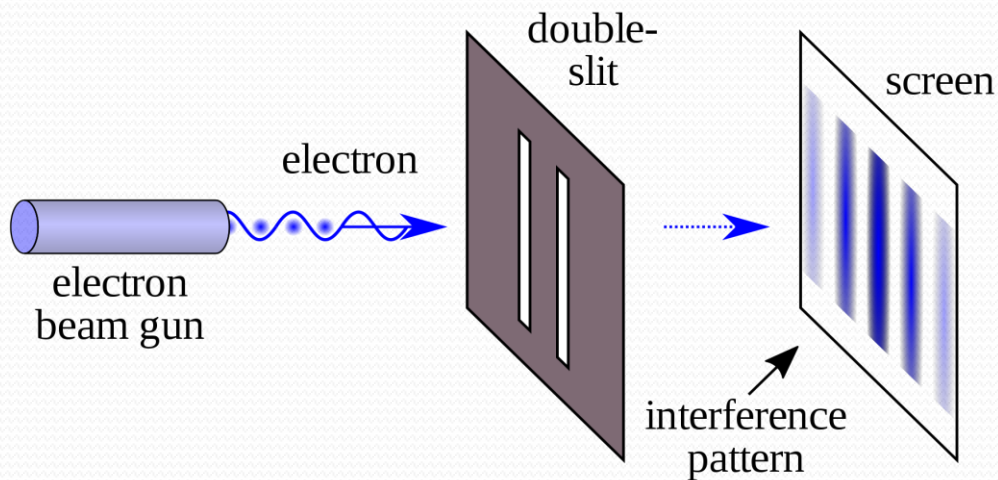


Creative interference gives sharpe bright points and destructive interference gives dark points



Dawison and Germer did Youngs double slit experiment for electrons in 1927.

- They observed same results as Young obtained in his experiment



- **So electrons are subject to interference.**



Electron Microscope

- ❑ Invented by German Scientist Ernest Ruska in 1931
- ❑ Earned Nobel Prize in 1986

Basic Principal,

For matter , $\lambda = h / \sqrt{2Em}$

When E increases λ decreases.

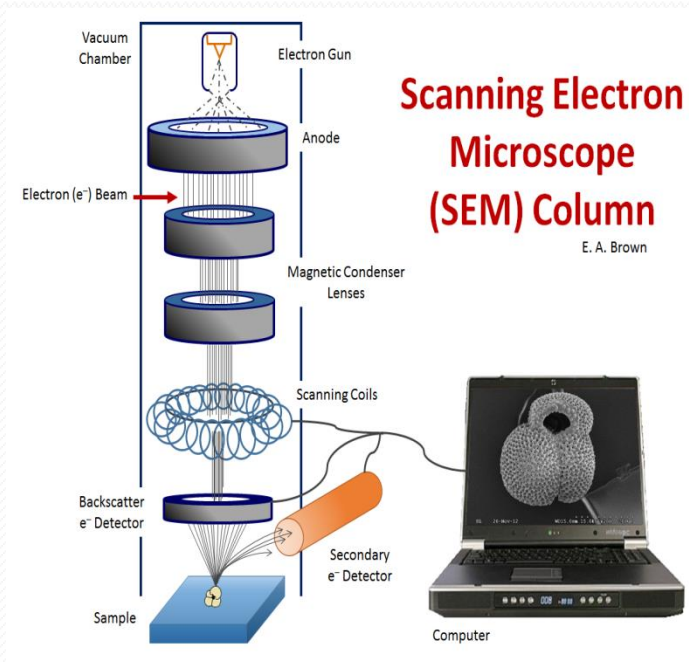
When λ minimizes that of light ,two points which are very closer can be separated & viewed clearly.
Hence Resolution power is high.





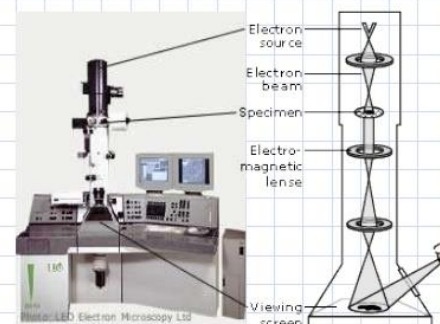
Types of Microscopes

- TEM-Transmission Electron Microscope
- SEM-Scanning Electron Microscope
- REM-Reflection Electron Microscope



Transmission Electron Microscopy

TEM is a unique tool in characterization of materials crystal structure and microstructure simultaneously by diffraction and imaging techniques.





Light Microscope & Electron Microscope



	Light Microscope	Electron Microscope
Magnification	1000-1500	10^5
Resolution	0.2 μm	0.5 nm
resolution	Light	Electron beam
Medium	Air	Vacuum
Lense	Optical Lense	Magnetic Field
Focusing	Movable lenses	Variable current to Magnets
Principal	Optics	Wave properties of electrons



Advantages

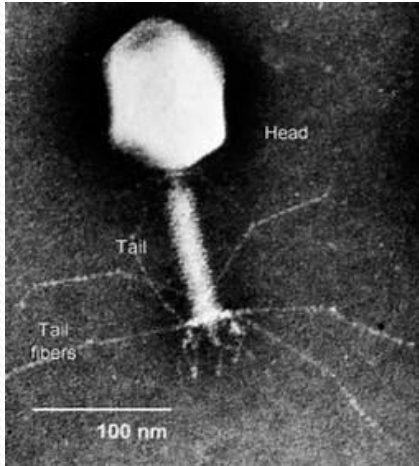
- Digital data to analyze easily
- High speed.
- Higher magnification & resolution.

Disadvantages

- Expensive
- Living specimens cannot be viewed.
- Need skilled workers.
- Size-harder to move & set up.
- Heat could deform specimens
- Need preparation of specimens



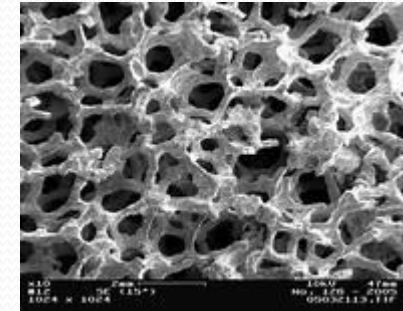
Images of macroscopic



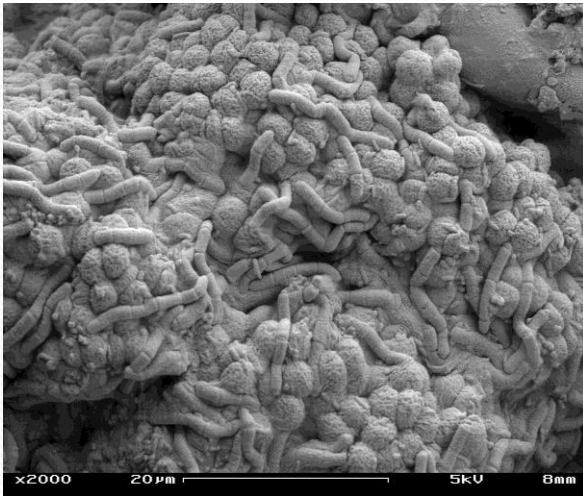
Virus



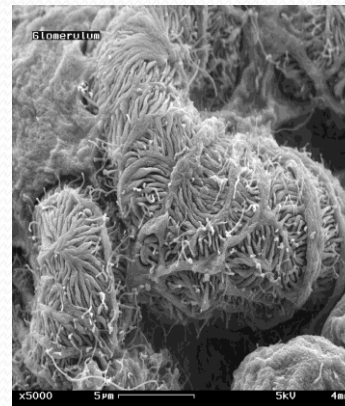
Cancer Cells



Metallic Structures



Algae & Bacteria



Glomerulum